

Factor Models to Measure Human Capital and Parental Investment

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1. Motivation

- We are interested in **human capital** and its **determinants** (e.g., family background, schooling environment, parental investment).
- Key constructs are often **latent**: they are difficult to observe directly and are not recorded as a single variable in most datasets.
- Available measures (test scores, grades, health indicators, time use, expenditures, attitudes) are:
 - **noisy** (measurement error),
 - **incomplete** (only some dimensions observed),
 - **not comparable** across contexts/ages without structure.
- Goal: build **credible, interpretable indices** of latent constructs and relate them to covariates and outcomes.

2. Why Factor Models?

- Factor models treat the construct of interest as a **latent variable** that drives multiple observed indicators.
- Intuition: we observe many measures of the same underlying thing, each with some error.
- Benefits:
 - ① **Separates signal from noise** (measurement error handled explicitly).
 - ② **Combines information** across multiple indicators into a smaller set of factors.
 - ③ Enables **comparisons** and **structural interpretation** (loadings, reliability).
 - ④ Natural extension to multiple factors (cognitive vs non-cognitive; time vs. monetary investments).

Key idea: Use covariance across indicators to infer the latent factor(s).

3. Example: Measuring Human Capital

Latent construct: Human capital H (multi-dimensional)

Possible observed indicators

- **Cognitive:** standardized test scores (math, reading), vocabulary tests, working memory tasks
- **Non-cognitive / socio-emotional:** grit, self-control, locus of control, social skills scales
- **Health / physical:** height-for-age, BMI z-score, hemoglobin, days ill, motor skills

Example measurement approach

- One-factor: $H \rightarrow$ (math, reading, GPA)
- Two-factor: $H^{(cog)}$ and $H^{(soc)}$ with correlated factors

4. Example: Measuring Parental Investment

Latent construct: Parental investment /

Possible observed indicators

- **Time inputs:** hours helping with homework, reading to child, joint activities
- **Material inputs:** education spending, books at home, tutoring, learning materials
- **Home environment:** stimulation index, parenting practices, routines, monitoring
- **School engagement:** meetings with teachers, participation in school events

Common modeling choices

- Distinguish **quantity** (time/material) vs **quality** (engagement/stimulation)
- Allow investments to vary by child age (measurement invariance across waves)

5. The Factor Model (Measurement + Structural)

Measurement model (one factor):

$$y_{ij} = \nu_j + \lambda_j \theta_i + \varepsilon_{ij}, \quad j = 1, \dots, J$$

- y_{ij} : observed indicator j for individual i
- θ_i : latent factor (e.g., human capital H_i or investment I_i)
- λ_j : loading (how strongly indicator j reflects the factor)
- ν_j : intercept baseline (expected) level of indicator j when the latent factor equals zero; absorbs scale and difficulty differences across indicators.
- ε_{ij} : measurement error (often mean 0, uncorrelated across j as a baseline)

Side note: indicator y_{ij} can be continuous, categorical or binary. Model can be extended to accomodate this.

6. Identification: What Pins Down the Scale?

Factor models are identified only up to **location** and **scale** without restrictions.

Common identification choices

- **Fix factor variance:** $\text{Var}(\eta) = 1$ and estimate all loadings
- **Fix one loading:** set $\lambda_1 = 1$ (reference indicator) and estimate $\text{Var}(\eta)$
- **Fix factor mean:** set $\mathbb{E}[\eta] = 0$ (often automatic in SEM software)

Additional considerations

- Need at least **3 indicators** per factor for stable estimation (rule of thumb).
- With multiple groups/waves, consider **measurement invariance**.

7a. Parental Investment: Measurement Model (lavaan)

```
library(lavaan)

# Measurement model: latent parental investment (I)
# Example indicators:
#   time_reading      = minutes/week reading to child
#   homework_help     = days/week helping with homework
#   edu_spending      = annual education spending
#   books_home        = number of children's books at home
#   school_meetings   = number of parent-teacher meetings
```


7b. Parental Investment: Identification choice 1 and Estimation

```
# Structural part: determinants of parental investment
model_I <- '
  # Measurement model (loadings free; scale set by Var(I)=1)
  I =~ NA*time_reading + homework_help + edu_spending +
      books_home + school_meetings

  # Fix the latent mean (intercept) to 0
  I ~ 0*1
',

# Estimate model
fit_I <- cfa(model_I, data = df, std.lv = TRUE)

summary(fit_I, fit.measures = TRUE)
```

7c. Factor Models: Identification choice 2 and Estimation

Fix first loading to 1 and its intercept to 0

```
library(lavaan)

model_I_ref <- '
  # Fix the first loading to 1 (reference indicator)
  I =~ 1*time_reading + homework_help + edu_spending +
      books_home + school_meetings

  # Fix intercept of reference indicator to 0
  time_reading ~ 0*1
',

# Identification via marker loading (no std.lv here)
fit2 <- cfa(model_I_ref, data = df)
```

Other examples and tutorials: <https://lavaan.ugent.be/tutorial/>

8. References

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